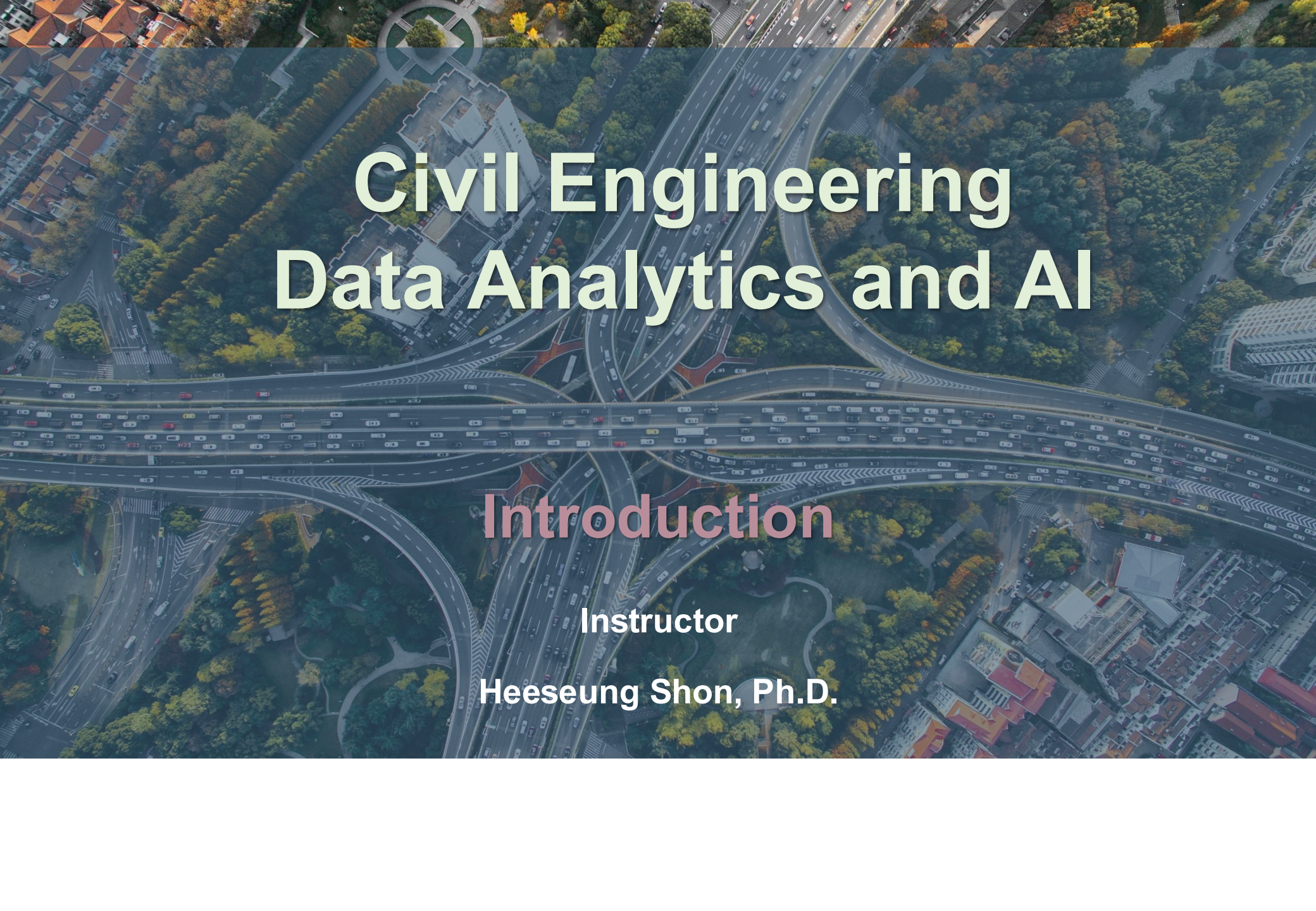




Civil Engineering Data Analytics and AI

Introduction

Instructor

Heeseung Shon, Ph.D.

Course Objectives

□ This course aims to develop students' knowledge and skills in:

1 Mathematical foundations and intuition for AI

- Understand **key mathematical concepts** behind data-driven models, including vectors and matrices, linear transformations, derivatives, and regularization.
- Explain how these concepts support **dimensionality reduction, model training,** and the **prevention of overfitting.**
- Use these ideas to **interpret model behavior and results** in civil engineering applications.

2 Modeling methods and model development

- Understand a **range of modeling methods**, including linear regression, decision trees and ensembles, support vector machines, k-means clustering, and neural networks.
- Select methods that are **appropriate** for the data, engineering problem, and analytical objective.
- Develop and compare models while addressing **overfitting, bias, and uncertainty** through **regularization and cross-validation.**

3 Data mining and reproducible practice with Python

- **Preprocess real-world data** by imputing missing values, handling outliers, and transforming variables.
- **Evaluate models** using appropriate metrics such as root-mean-square error, precision, and recall.
- Implement, document, and communicate a **reproducible data-analysis workflow in Python.**



Goal: Use data and AI responsibly to solve civil engineering problems and create value for society.

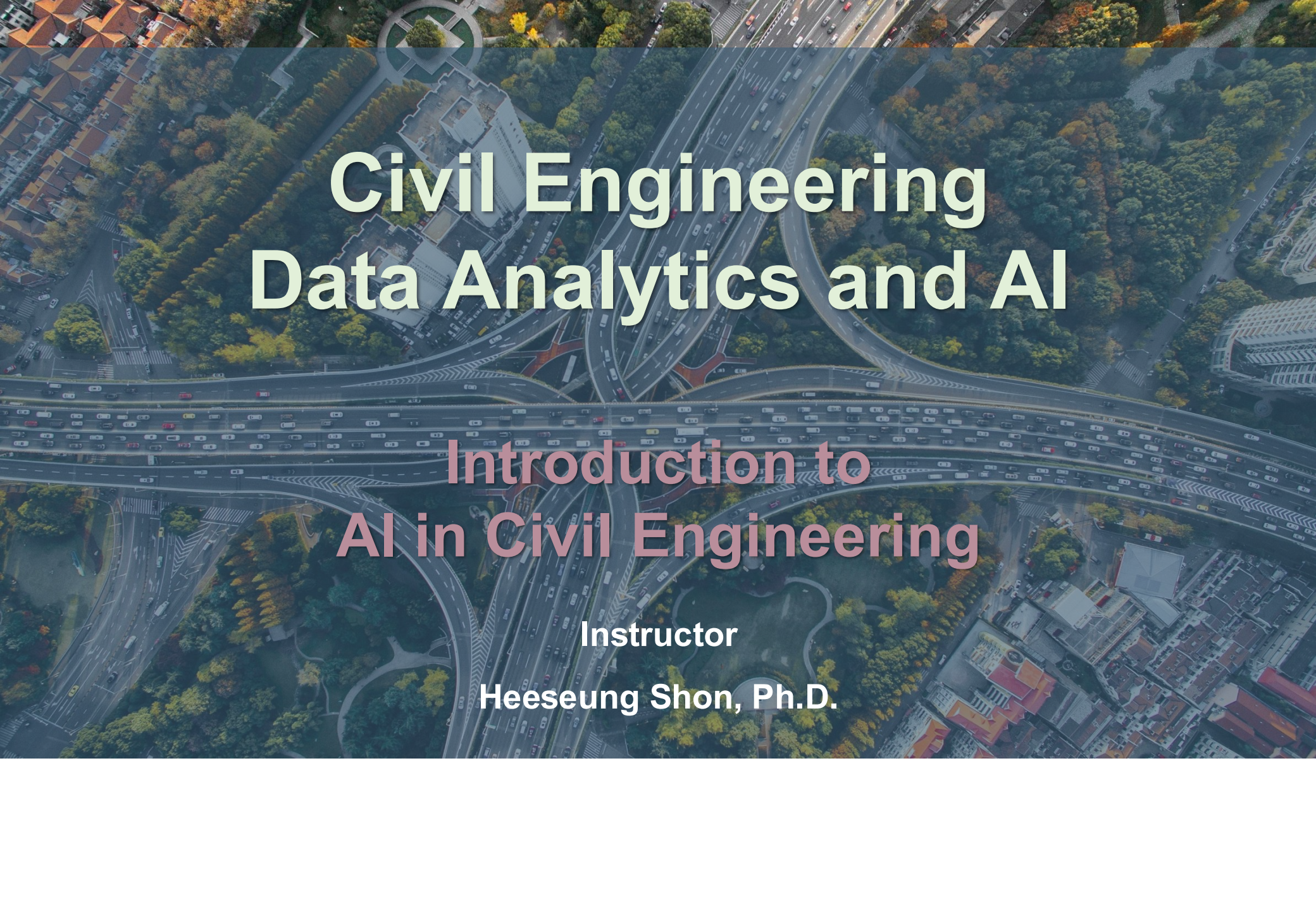


| Textbook and References

- ❑ **Lecture Note**
- ❑ **Hands-On Machine Learning with Scikit-Learn,
Keras & TensorFlow (Aurelien Geron)**
- ❑ **Deep Learning (Goodfellow, Bengio, Courville)**

Class Contents

Introduction to AI in Civil Engineering
Data Preparation and Exploratory Analysis
Applied Math and Machine Learning Basics
Regression Models
Support Vector Machines
Decision Trees and Ensemble Learning
Clustering
Dimensionality Reduction
Neural Networks and Deep Learning
CNNs, RNNs, and GNNs
Model Evaluation and Generalization



Civil Engineering Data Analytics and AI

Introduction to AI in Civil Engineering

Instructor

Heeseung Shon, Ph.D.

Introduction to AI in Civil Engineering

Smart Cities and Digital Twins

□ Smart Cities

- A smart city uses **IoT sensors** and technology to **connect** components across a city to derive **data** and improve the lives of citizens and visitors.



Introduction to AI in Civil Engineering

Smart Cities and Digital Twins

❑ Why Smart Cities Matter to Civil Engineers

- A smart city is a system of interconnected infrastructures.
- Civil engineers make important decisions across these systems.

❑ Examples



Transportation

- Traffic signal control
- Congestion management
- Travel demand and departure-time management



Energy Supply

- Energy generation planning
- Power plant siting



Water Supply and Treatment

- Water level control
- Treatment operations
- Dam location

Introduction to AI in Civil Engineering

Smart Cities and Digital Twins

❑ Digital Twins in Civil Engineering

- A digital twin is **a dynamic virtual representation** of a physical system, continuously updated with real-world data.



Physical world



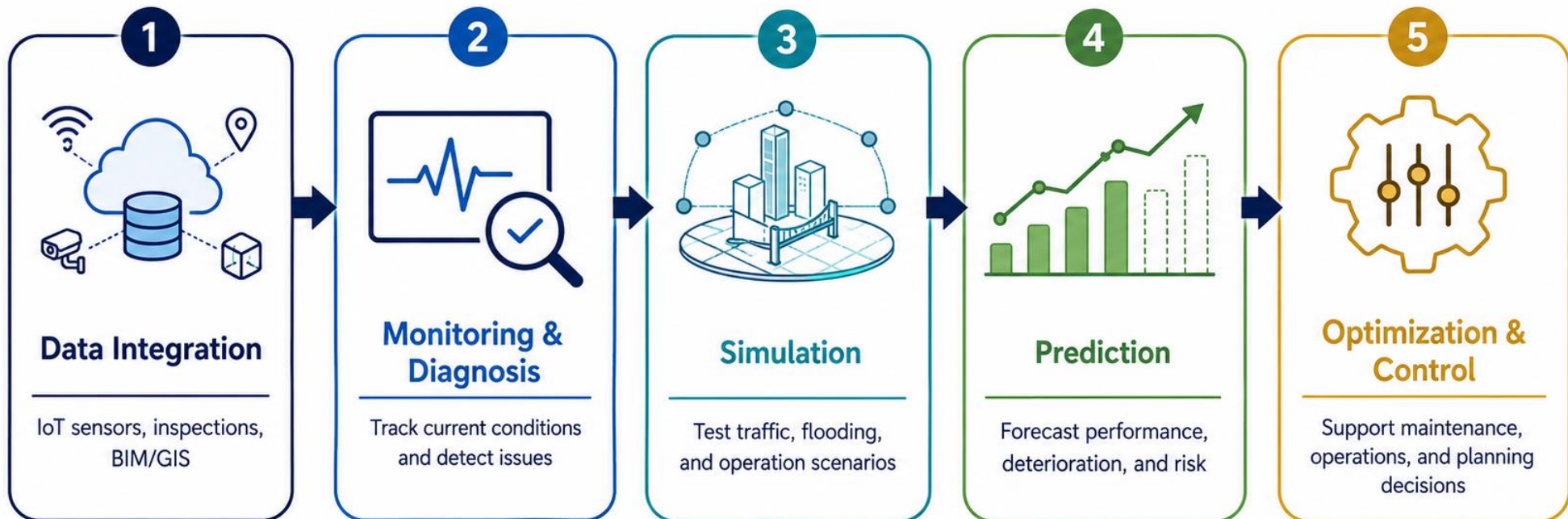
Digital twin

Introduction to AI in Civil Engineering

Smart Cities and Digital Twins

□ Digital Twins in Civil Engineering

- Digital twins are used throughout the **infrastructure lifecycle** to **simulate performance**, **predict future conditions**, and **optimize engineering decisions**.

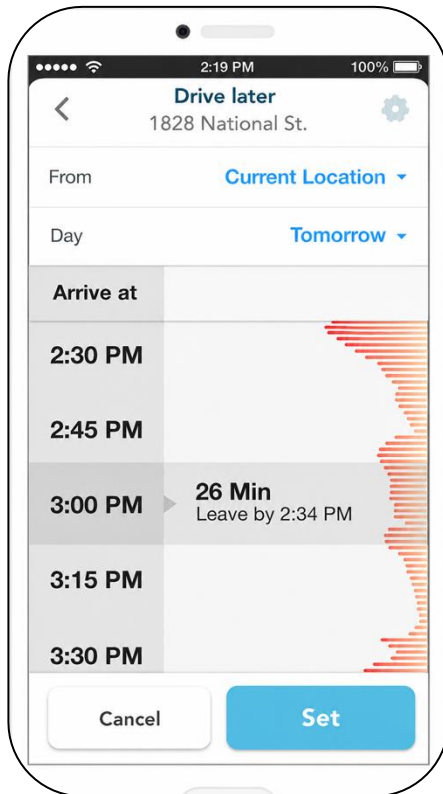


Introduction to AI in Civil Engineering

AI in Transportation Engineering

❑ Example: Travel Time Prediction

Driving travel-time prediction



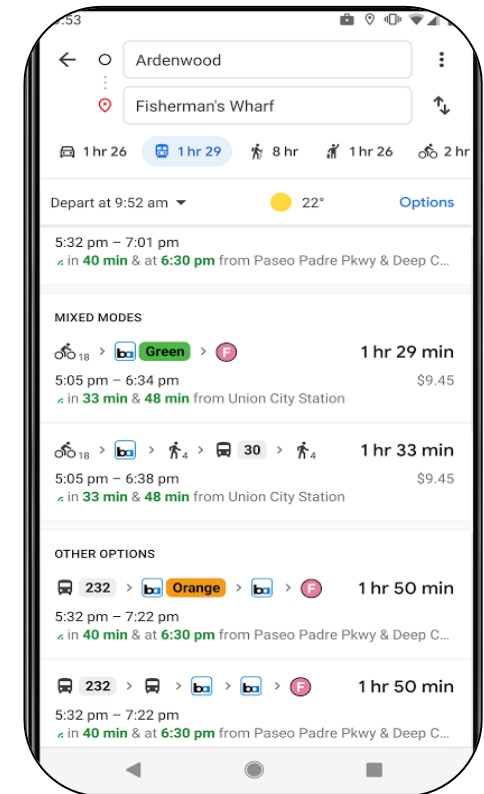
Ex) Waze, Google Maps

Transit arrival-time prediction



Ex) NJ Transit

Multimodal travel-time prediction



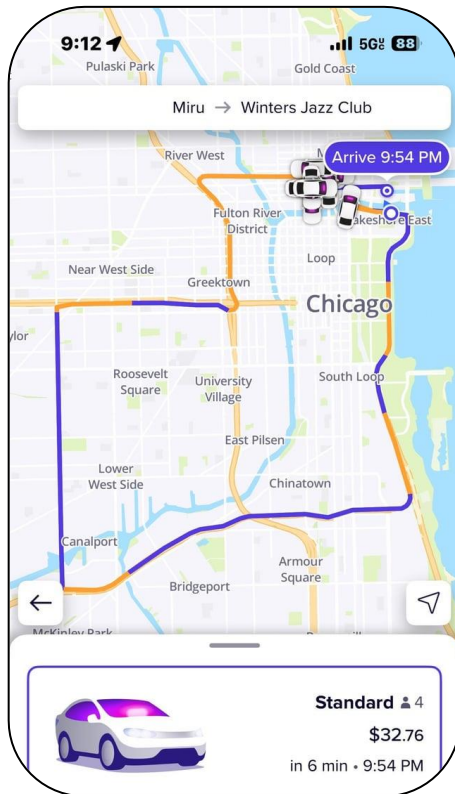
Ex) Google Maps

Introduction to AI in Civil Engineering

AI in Transportation Engineering

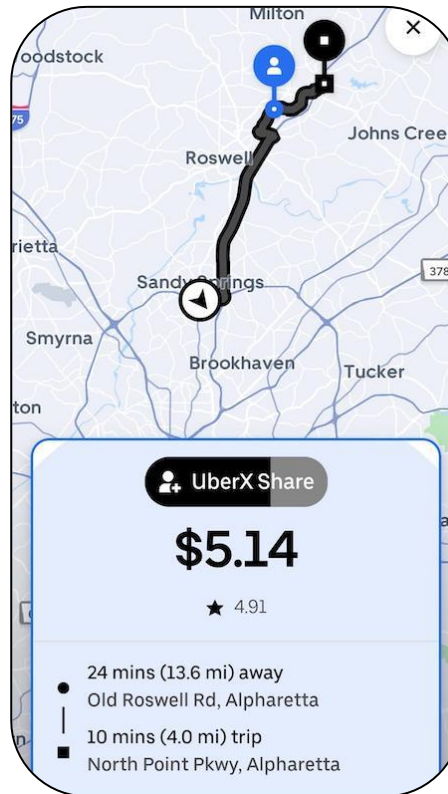
□ Example: AI-Enhanced Route Optimization

Individual route optimization



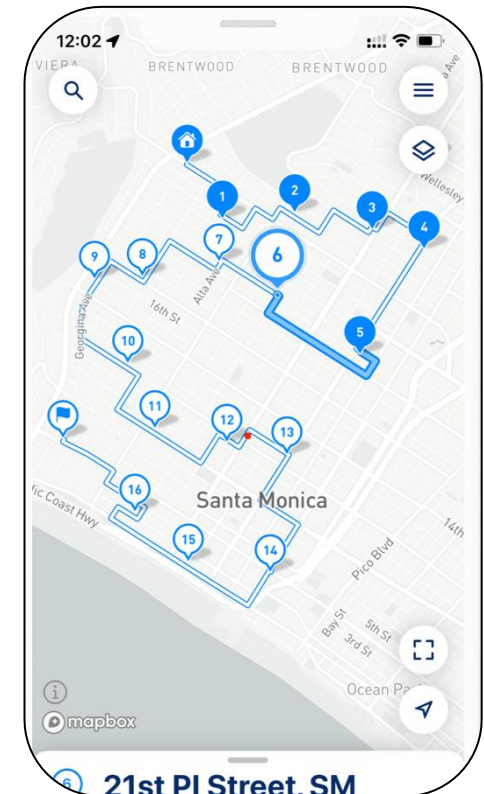
Ex) Uber, Lyft

Ridesharing route optimization



Ex) Uber, Lyft

Delivery route optimization



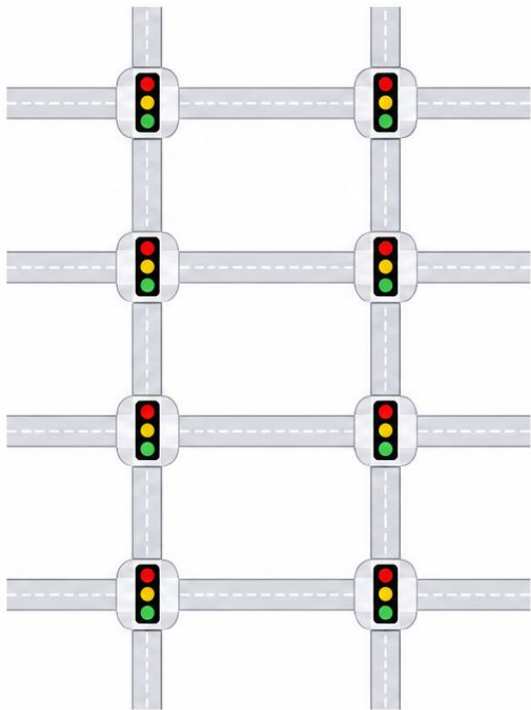
Ex) Amazon, Walmart

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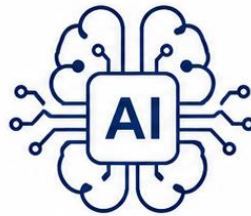
AI in Transportation Engineering

□ Example: AI-Enhanced Signal Control

Too Many Combinatorial Cases



AI



Multi-agent
planning



Automated
scheduling



Decentralized
decision-making

Pittsburgh SURTRAC

AI-based decentralized
adaptive signal control



Travel time ↓ 17-33%



Wait time ↓ 28-50%



Stops ↓ 9-53%

Introduction to AI in Civil Engineering

Infrastructure Monitoring and Predictive Maintenance

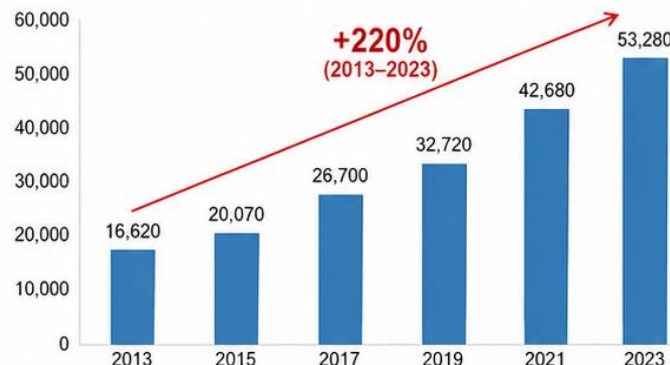
❑ Road Pavement Monitoring



Pothole-Related Crashes and Social Costs in the U.S.



Pothole-Related Crashes Are Increasing (Number of Crashes)



Social Costs Due to Pothole Crashes Are Increasing

(Social Cost, Million USD)



Introduction to AI in Civil Engineering

Infrastructure Monitoring and Predictive Maintenance

Image-based Road Defect Detection Technologies

1 UAV image-based road defect detection (Joo, 2017)

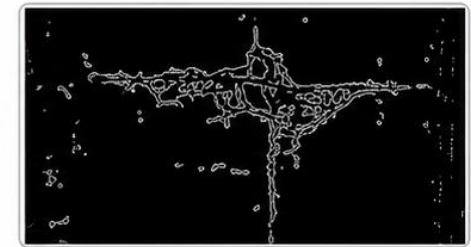
- Using region-of-interest setting and Canny edge detection (accuracy: 91%).



1 Aerial image with ROI



2 Extracted defect region (crack overlay)



3 Canny edge detection result

2 Deep learning-based pothole detection (Hwang et al., 2021)

- YOLO-based pothole detection (accuracy: 98%).



1 Video recording



2 Datasets



3 Training

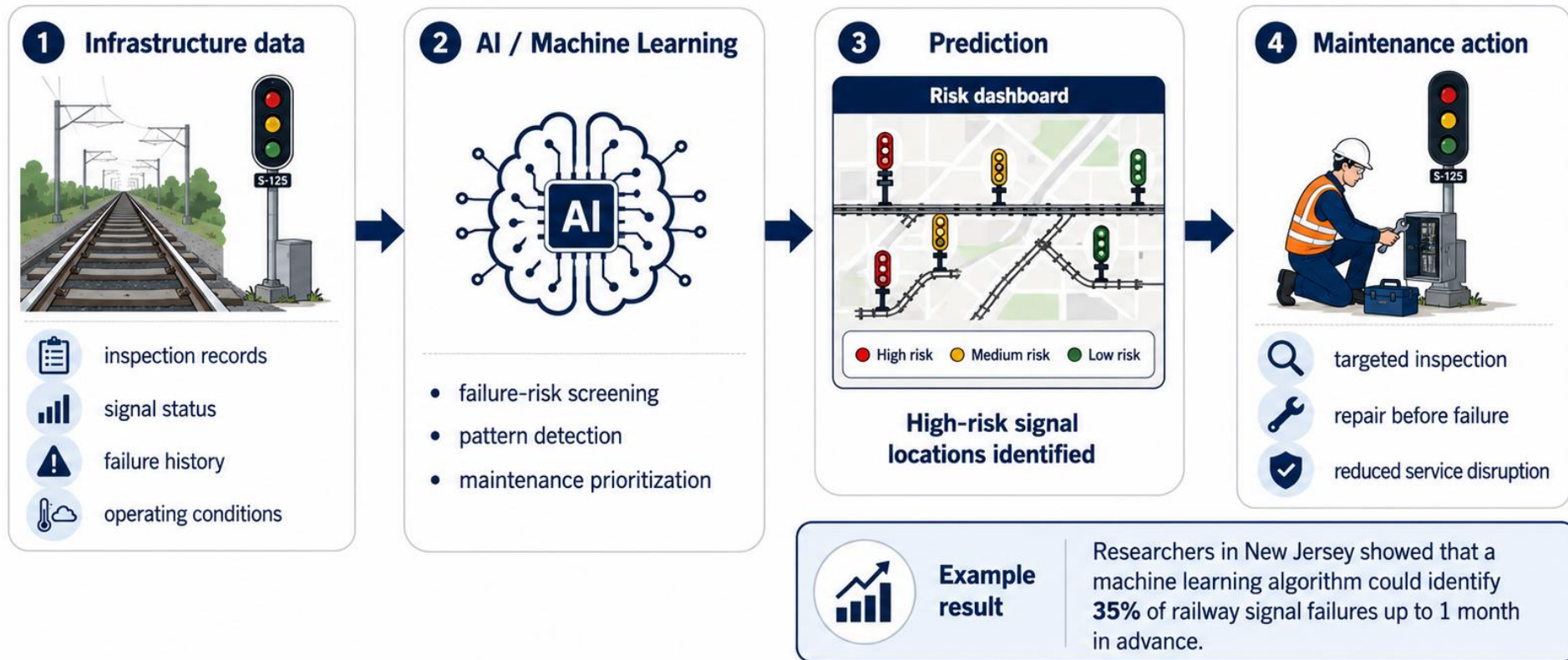


4 Detection

Introduction to AI in Civil Engineering

Infrastructure Monitoring and Predictive Maintenance

□ Railway Signal Predictive Maintenance



Predict failures earlier → prioritize maintenance → improve reliability and safety

Introduction to AI in Civil Engineering

Construction Automation and Safety

❑ Computer Vision for Construction

